

6.1 Genetics and evolution

6.1.1 Cellular control

The way in which cells control metabolic reactions determines how organisms, grow, develop and function.

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) types of gene mutations and their possible effects on protein production and function	To include substitution, insertion or deletion of one or more nucleotides AND the possible effects of these gene mutations (i.e. beneficial, neutral or harmful).

(b) the regulatory mechanisms that control gene expression at the transcriptional level, post-transcriptional level and post-translational level

To include control at the:

- transcriptional level: *lac* operon, and the general role of transcription factors in eukaryotes (Learners are not required to recall specific transcription factors)
- post-transcriptional level: the editing of primary mRNA and the removal of introns to produce mature mRNA
- post-translational level: the activation of proteins by cyclic AMP.

HSW2

(c) the genetic control of the development of body plans in different organisms

The Homeobox gene sequences are similar and highly conserved in plants, animals and fungi, and are involved in regulating gene expression

AND

the role of Hox genes (as a subset of Homeobox genes) in controlling body plan development.

HSW7

(d) the importance of mitosis and apoptosis as mechanisms controlling the development of body form.

To include an appreciation that the genes which regulate the cell cycle and apoptosis are able to respond to internal and external cell stimuli.